

Causal Rigidity and the Single-Unit Universe: Integrating the Alexandrov-Zeeman and Unruh clock scales

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We unify two complementary viewpoints on relativistic spacetime and the counting of fundamental constants. Operationally, Matsas, Pleitez, Saa, and Vanzella (MPSV) have recently argued that relativistic spacetime requires only a single fundamental dimensional constant, demonstrating this via a clock-based protocol that measures spatial distance without light rays. Mathematically, theorems due to Alexandrov and Zeeman demonstrate that the light-cone structure determines the spacetime geometry only up to a conformal factor. We show that these approaches are mutually reinforcing: the Alexandrov-Zeeman theorems establish the rigid conformal structure of spacetime, while the “bona fide clock” required by MPSV serves the necessary mathematical role of breaking the dilation symmetry. We provide a formal derivation proving that the normalization of a single clock worldline is sufficient to select a unique metric from the conformal class, thereby clarifying that the number of fundamental constants is exactly one.

I. INTRODUCTION

The question “How many fundamental dimensional constants does physics require?” has been a subject of enduring debate, most famously in the “trialogue” between Duff, Okun, and Veneziano (DOV) [1]. While there is consensus that only dimensionless quantities are ultimately observable, disagreements persist regarding the number of independent *standards* (or base units) necessary to describe dimensionful observables.

Recently, Matsas, Pleitez, Saa, and Vanzella (MPSV) [2] revisited this issue from a *spacetime-based* perspective. They argue that once a relativistic spacetime is assumed as the starting point, the units required to construct that spacetime are sufficient to express all observables. Specifically, they conclude that in a relativistic spacetime, the number of fundamental dimensional constants is *one*.

Parallel to this metrological debate, the mathematical relativity community has long established—via theorems by Alexandrov [3] and Zeeman [4]—that the null (light-cone) structure of spacetime is incredibly rigid. These theorems state that any map preserving the causal order is a Lorentz transformation up to a global dilation (scale factor).

In this paper, we unify these two perspectives. In Sec. II, we review the operational argument provided by MPSV. In Sec. III, we review the geometric constraints imposed by causal automorphisms. Finally, in Sec. IV, we provide a formal derivation demonstrating that the “bona fide clock” of MPSV is the precise mathematical object required to break the conformal symmetry left open by incidence geometry.

II. THE OPERATIONAL PERSPECTIVE: MEASURING SPACE WITH TIME

MPSV argue that to construct a relativistic spacetime physically, one requires specific apparatuses. Unlike Galilean spacetime, which requires both rulers and clocks (because space and time are decoupled), relativistic spacetime requires only “bona fide clocks.”

To prove that spatial rulers are redundant, MPSV utilize a protocol attributed to William George Unruh. Consider three inertial clocks C_1, C_2, C_3 :

1. C_1 travels from an observer A to B .
2. C_2 travels from B back to A .
3. C_3 remains at rest with A .

If τ_1 and τ_2 are the proper time durations measured by the traveling clocks, and τ_3 is the duration measured by the stationary clock, the spatial distance D between A and B is given by Heron’s formula:

$$D = \frac{\sqrt{(\tau_3^2 - \tau_1^2 - \tau_2^2)^2 - 4\tau_1^2\tau_2^2}}{2\tau_3}. \quad (1)$$

This formula yields a spatial length in units of time (seconds). It demonstrates physically that the spatial metric can be derived entirely from time measurements, rendering the meter a “disposable” unit and c a mere conversion factor.

III. THE GEOMETRIC PERSPECTIVE: ALEXANDROV-ZEEMAN THEOREMS

We now turn to the mathematical structure of Minkowski spacetime. We consider the manifold $M \simeq \mathbb{R}^n$ (with $n \geq 3$), equipped with the standard flat metric $\eta_{ab} = \text{diag}(-1, 1, \dots, 1)$. The defining feature of relativistic geometry is the causal structure, encoded either

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by the incidence of light cones or the partial causal ordering of events.

A fundamental question in mathematical relativity is the extent to which this causal structure alone determines the geometry. This is resolved by a class of rigidity results known as Alexandrov-Zeeman type theorems.

A. Alexandrov: The Rigidity of Light Cones

In the 1950s, A. D. Alexandrov proved that the geometry of spacetime is encoded in the “skeleton” of light rays [3]. Let C_x denote the light cone (null cone) at an event $x \in M$.

Theorem 1 (Alexandrov). *Let $f : M \rightarrow M$ be a bijection of the spacetime manifold ($n \geq 3$). If f maps null lines to null lines (i.e., preserves the light-cone structure), then f is a linear conformal automorphism.*

This result implies that mere incidence—knowing which events can be connected by a light ray—fixes the linear structure of spacetime [5].

B. Zeeman: Causality Implies the Lorentz Group

In 1964, E. C. Zeeman strengthened this perspective by focusing on the partial ordering of events [4]. We say $x \prec y$ if a signal can travel from x to y (with speed $v \leq c$).

Theorem 2 (Zeeman). *Let $f : M \rightarrow M$ be a bijection that preserves the causal order (i.e., $x \prec y \iff f(x) \prec f(y)$). Then f is generated by:*

1. An orthochronous Lorentz transformation $\Lambda \in O^\uparrow(1, n-1)$,
2. A spacetime translation a , and
3. A global dilation (scaling) $x \mapsto \lambda x$ with $\lambda > 0$.

C. The Conformal Ambiguity

Both theorems lead to the same physical conclusion: the causal structure determines the metric g_{ab} up to a conformal factor $\Omega(x)$. Under the strict conditions of mapping the whole of Minkowski space to itself, this factor becomes a global constant, $\Omega(x) = \lambda$.

$$g'_{ab} = \lambda^2 \eta_{ab}. \quad (2)$$

This leads to a profound physical consequence: **A universe defined solely by light rays (or causal order) cannot distinguish a second from a millennium.** The transformation $t \rightarrow 1000t$ is a valid symmetry of the causal structure. The “shape” of physics is preserved, but the duration is not. To distinguish time scales, one must break this conformal symmetry.

IV. FORMAL UNIFICATION: THE CLOCK AS SYMMETRY BREAKER

We now formalize the connection between the MPSV clock and the Alexandrov-Zeeman conformal factor. We show that specifying the normalization of a single clock worldline is sufficient to fix the global scale λ .

A. Definitions

Definition 1 (Conformal Class). *Let η be the standard Minkowski metric. The conformal class $[\eta]$ determined by the causal structure is the set of metrics $\{\lambda^2 \eta : \lambda > 0\}$.*

Definition 2 (Bona Fide Clock). *Let $\gamma : \mathbb{R} \rightarrow M$ be a smooth, future-directed timelike worldline of the clock. A bona fide clock is the pair (γ, τ) where τ is a parameter on the domain $\mathbb{R}$ such that, with respect to the true physical metric $g_\star$, the parameter equals the proper time. That is, for all $s \in \mathbb{R}$:*

$$g_\star(\dot{\gamma}(s), \dot{\gamma}(s)) = -1, \quad (3)$$

where dot denotes d/ds .

Physically, this condition enforces that the parameter τ is synchronized with the spacetime geometry. It asserts that the tangent vector $\dot{\gamma}$, which represents the clock’s velocity through spacetime, has a constant, unit magnitude. In other words, for every “tick” of the parameter s , the system traverses exactly one unit of invariant interval defined by $g_\star$. This is what elevates s from an arbitrary coordinate label to a physical measure of duration, thereby fixing the scale of the metric.

B. Uniqueness Theorem

Theorem 3 (Single Clock Fixes Scale). *Let $M \cong \mathbb{R}^4$ be equipped with a causal order determining the conformal class $[\eta]$. Let (γ, τ) be a single bona fide clock following an inertial trajectory. Then, the requirement that τ measures proper time with respect to the physical metric $g_\star \in [\eta]$ uniquely determines $g_\star$.*

Proof. Since the causal structure fixes the metric up to a global scale (via the Alexandrov-Zeeman theorems for $n \geq 3$), the physical metric must take the form:

$$g_\star = \lambda^2 \eta, \quad \lambda > 0, \quad (4)$$

where η is an arbitrary reference metric (e.g., standard Minkowski coordinates).

The bona fide clock condition (Eq. 3) requires that along the worldline γ :

$$g_\star(\dot{\gamma}, \dot{\gamma}) = -1 \text{ which implies } \lambda^2 \eta(\dot{\gamma}, \dot{\gamma}) = -1. \quad (5)$$

Let $-\alpha^2 \equiv \eta(\dot{\gamma}, \dot{\gamma})$ be the squared norm of the clock’s tangent vector as measured in the reference metric η . Since

γ is timelike, $\alpha^2 > 0$. This quantity α physically encodes the “tick rate” of the clock relative to the arbitrary reference coordinates. It represents the real, positive magnitude of the clock’s velocity vector through the reference background η , quantifying the mismatch between the coordinate measuring system and the physical clock.

The equation becomes:

$$\lambda^2(-\alpha^2) = -1 \implies \lambda = \frac{1}{\alpha}. \quad (6)$$

Since α is a determined value based on the coordinate description of the curve in the reference system, λ is uniquely determined. Thus, $g_\star$ is unique. $\square$

C. The Hierarchy of Metrics

It is instructive to explicitly distinguish the roles of the metric tensors discussed above. The relationship between the generic conformal scaling and the specific solution $g_\star$ can be summarized as follows:

1. **The Reference Metric (η_{ab}):** An arbitrary mathematical auxiliary (the standard Minkowski metric) that correctly describes the causal incidence (light cones) but possesses no physical scale.
2. **The Conformal Class (g'_{ab}):** The set of all possible metrics allowed by the Alexandrov-Zeeman theorems. This is a one-parameter family generated by scaling the reference metric:

$$g'_{ab} = \lambda^2 \eta_{ab}, \quad \forall \lambda \in (0, \infty). \quad (7)$$

At this level, the theory cannot distinguish between a second and a millennium; any value of λ preserves the causal order.

3. **The Physical Metric ($g_\star$):** The unique element of the conformal class that is selected by the physical apparatus.

Mathematically, $g_\star$ is obtained by evaluating the general form g'_{ab} at the specific scale factor λ_{phys} determined by the clock normalization:

$$g_\star \equiv g'_{ab}|_{\lambda=1/\alpha} = \frac{1}{\alpha^2} \eta_{ab}. \quad (8)$$

Thus, while the causal structure generates the candidate list defined by η_{ab} , the single bona fide clock collapses this list to the unique physical reality $g_\star$.

V. DISCUSSION

The formal derivation above confirms the operational intuition of MPSV. The “one fundamental constant” is the physical realization of the mathematical mechanism required to break the dilation symmetry of incidence geometry.

- *Incidence Relations (Alexandrov/Zeeman):* These fix the conformal class $[\eta]$. Physically, they define the “shape” of spacetime and the structural role of c , but leave the scale undetermined.
- *The Clock:* This provides the normalization condition $g(\dot{\gamma}, \dot{\gamma}) = -1$, eliminating the dilation sector.

Therefore, the counting of fundamental constants in Special Relativity is exactly one because the symmetry group of the underlying structure is the Poincaré group extended by dilations. The causal structure provides the Poincaré sector, and the single unit (the clock) fixes the scale.

Finally, regarding the role of c : kinematically, MPSV [2] correctly identify c as a mere unit converter between the fundamental time and the derivative space. Yet, dynamically, c retains a privileged status. It is the unique velocity parameter that ensures the form invariance of the equations of motion, such as those of electrodynamics [6, 7]. Thus, while the clock fixes the scale of the universe, c defines the rigid phenomenal structure necessary for the clock to operationally tick in such a way as to guarantee the form invariance of the physical laws under Lorentz boosts.

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